DEEP LEARNING FINAL PROJECT - MULTI-RETINAL DISEASE DETECTION

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1. INTRODUCTION

Even though deep learning has shown strong results in medical image analysis, its performance often depends on large, labeled datasets. These challenges include the need for expert knowledge, privacy concerns that limit data sharing, and the expense of collecting high-quality annotations, all of which are common in medical imaging. The use of retinal fundus images has become popular for screening for eye diseases such as Diabetic Retinopathy (DR), Glaucoma(G), and Age-related Macular Degeneration (AMD). Hence, training these models from scratch can easily lead to overfitting and poor generalization. In such cases, transfer learning is a practical solution, where models pretrained on large datasets are reused and applied to smaller target datasets to improve stability and performance.

The goal of this project was to improve retinal multi-label classification performance using transfer learning, while also developing insight into what helps when data is limited. Also, this evaluates the results off-site in detail, along with an on-site result via Kaggle.

2. METHODS

The project came in four tasks to complete.

In task 01, the ODIR retinal fundus dataset was used to test two pretrained backbones: EfficientNet and ResNet18. These were pretrained on large retinal datasets and then adapted to ODIR. First, the pretrained model was evaluated directly on ODIR with no fine-tuning. Secondly, the backbone was frozen, and only the classifier was updated using the ODIR training set. Lastly, there was a full fine-tuning, where both the backbone and classifier were updated using the training set.

Since DR cases were more frequent than G and AMD cases in the training data, task 02 focused on class imbalance and its mitigation. Two loss functions were implemented and compared using ResNet18. This was done in two steps: first, the dominance of easy examples was reduced by emphasizing harder misclassified samples. Then, BCE was reweighted by class frequency to strengthen minority-class learning.

The third task investigated whether adding attention improved feature focus for disease recognition. Two attention modules were implemented and integrated into the ResNet18.

The fourth and last task was to explore additional strategies to further improve performance. Among the suggested methods, the Swin Transformer and Vision Transformer models were not selected despite their potential to improve the F1-score, because they required full retraining across all epochs, which could result in long training times. Grad-CAM-based attention retraining was excluded due to its multi-stage pipeline, lengthier execution time, and high implementation complexity. Variational Autoencoders were also rejected due to the need for extensive training and validation and the associated risks. Hence, the Weighted Average Ensemble (WAE) was implemented, combining multiple models and exploiting model diversity to reduce errors while remaining efficient, simple, feasible, and short on execution time.

4. RESULTS

4.1. Task 01

- **Task 1.1**

Offsite-Results

The two pretrained backbones were evaluated without any fine-tuning.

EfficientNet: Average F1-score 54.7%

	DR	Glaucoma	AMD
Precision	0.7360	0.5536	0.2278
Recall	0.6571	0.6327	0.8182
F1-score	0.6943	0.5905	0.3564

ResNet18: Average F1-score 52%

	DR	Glaucoma	AMD
Precision	0.7238	0.5625	0.2537
Recall	0.5429	0.5510	0.7727
F1-score	0.6204	0.5567	0.3820

Onsite- Results

EfficientNet	0.60471	ResNet18	0.56920

As for the results, EfficientNet achieved higher baseline performance than ResNet18, indicating that the EfficientNet backbone generalized better to the unseen onsite data.

- **Task 1.2**

Offsite – Results

A frozen-backbone fine-tuning strategy was applied, in which the feature extractor weights were kept fixed, and only the final classifier layer was optimized.

EfficientNet: Average F1-score 63.5%

	DR	Glaucoma	AMD
Precision	0.8361	0.7500	0.4444
Recall	0.7286	0.5510	0.5455
F1-score	0.7786	0.6353	0.4898

ResNet18: Average F1-score 54.0%

	DR	Glaucoma	AMD
Precision	0.7805	0.6364	0.8182
Recall	0.9143	0.1429	0.4091
F1-score	0.8421	0.2333	0.5455

Onsite- Results

EfficientNet	0.74243	ResNet18	0.61263

EfficientNet was chosen as the better performer here as it has a substantially higher score, even though ResNet18 gives the best value within the range.

- **Task 1.3**

EfficientNet: Average F1-score 71.4%

	DR	Glaucoma	AMD
Precision	0.8643	0.7209	0.6190
recall	0.8643	0.6327	0.5909
F1-score	0.8643	0.6739	0.6047

ResNet18: Average F1-score 75.5%

	DR	Glaucoma	AMD
Precision	0.9328	0.8085	0.7059
Recall	0.7929	0.7929	0.5455
F1-score	0.8571	0.8571	0.6154

Onsite- Results

EfficientNet	0.79615	ResNet18	0.77831

The ResNet18 results were more closely aligned with the reference range, whereas the EfficientNet results were lower than the corresponding reference mean.

4.2 Task 02

- **Task 2.1**

Offsite – Results

A focal loss-based training setup was applied to reduce the dominance of the majority DR class by down-weighting easy examples and emphasizing misclassified examples.

ResNet18: Average F1-score 73.8%

	DR	Glaucoma	AMD
Precision	0.8968	0.7551	0.5833
Recall	0.8071	0.7551	0.6364
F1-score	0.8496	0.7551	0.6087

Onsite- Results

ResNet18	0.77468

With the results, ResNet1 performed better as a backbone.

- **Task 2.2**

Offsite – Results

A class-balanced reweighting scheme was applied to the BCE loss to reduce the impact of frequency imbalance across others.

ResNet18: Average F1-score 75.0%

	DR	Glaucoma	AMD
Precision	0.8992	0.8085	0.5600
Recall	0.8286	0.7755	0.6364
F1-score	0.8625	0.7917	0.5957

Onsite- Results

ResNet18	0.80604

4.3 Task 03

- **Task 3.1**

Offsite – Results

A ‘squeeze and excitation’ (SE) module was incorporated into the ResNet18 backbone, and then the performance was evaluated.

ResNet18: Average F1-score 74.1%

	DR	Glaucoma	AMD
Precision	0.9024	0.7600	0.5556
Recall	0.7929	0.7755	0.6818
F1-score	0.8441	0.7677	0.6122

Onsite- Results

ResNet18	0.80456

- **Task 3.2**

Multi-Head Attention (MHA) was integrated to let the model attend to multiple complementary feature subspaces simultaneously.

Offsite – Results

ResNet18: Average F1-score 74.1%

	DR	Glaucoma	AMD
Precision	0.9024	0.8085	0.5769
Recall	0.7929	0.7755	0.6818
F1-score	0.8441	0.7917	0.6250

Onsite- Results

ResNet18	0.80550

4.4 Task 04

Using the WAE method improved the F1-score's predictive efficiency to 0.79591, recording 1% improvement.

Model	Average F1 score
EfficiNet	77.4%
ResNet18	78.4%
Ensemble	79.5%~80%

5. DISCUSSION AND CONCLUSION

In task 1.1 off-site, DR showed the strongest performance, while AMD showed the weakest. For AMD, high recall and low precision were observed, indicating that its positives and limited class separation occurred without fine-tuning. Even though minor deviations were attributed to implementation differences, onsite scores closely matched the reference, confirming that the evaluations and submissions were accurate. Overall, EfficientNet performed better here. This method provides a proper baseline using pretrained weights. However, the class separation is limited, leading to instability. Across both offsite and onsite evaluations, Task 1.2 showed a disease-dependent result: offsite performance detected DR reliably, while Glaucoma and AMD remained challenging, indicating that training only the classifier can limit adaptation. On-site set, achieved a F1-score of 0.61263 with ResNet18, within the reference range, supporting the implementation, inference procedure, and submissions. This method's performance improvement was done with minimal computing, yet the harder classes were not consistently learned. In task 1.3, the method enabled more task-specific features and stronger generalization, yet training was prone to overfitting.

For task 2.1, focal loss was applied to address the severe class imbalance, resulting in consistent improvements. Across both evaluations. On both the on-site and off-site sets, ResNet18 outperformed EfficientNet, indicating that focal loss helped reduce bias towards the majority class and improved generalization, while ResNet18 provided the most stable performance. This method improved minority-class sensitivity, but precision was sometimes dropped. Class-balanced reweighting of the BCE loss was applied to mitigate the strong class-frequency imbalance in task 2.2, resulting in stable improvement with ResNet18. Overall results indicated that the class-balanced loss improved minority-class learning without sacrificing majority-class performance, leading to robust generalization on the unseen onsite data. The incorporation of the squeeze-and-excitation module in task 3.1 led to strong, consistent performance across both off-site and on-site evaluations. This is believed to enhance feature representation and generalization while producing reliable performance gains without introducing instability across classes. This method emphasized informative features yet it added more complexity. Using the method from 3.2, strong and balanced performance was observed across both splits. This module appeared to enhance representational flexibility and generalization. The WAE achieved the best average of 79.5%-80%, improving by about 1% over the single-model baselines, indicating that combining the models with tuned thresholds improved prediction consistency on the DR-dominant dataset. On-site tests supported the results obtained.

6. REFERENCES

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[2] **Kuncheva, L. I. (2004).** *Combining Pattern Classifiers: Methods and Algorithms.* Wiley-Interscience.

Team Statement

Aloka: Tasks 2 and 3, and the results and discussion section for the said tasks.

Thiranja: Task 1, Task 5 (writing and editing the whole report, the introduction, and the methods.

Imindu: Task 4, results, and discussion related to the task.